So far we have estimated limits using tables and graphs. The following examples show that these methods can be very unreliable.

Example 1

1. Find $\lim_{x\to 0}\cos(\frac{1}{x})$. Here is a table to help you:

x	$\cos(\frac{1}{x})$
± 0.01	0.86232
± 0.001	0.56238
± 0.0001	-0.95216
± 0.00001	-0.99936

   $\lim_{x\to 0}\cos(\frac{1}{x}) =$

2. Fill in the next few entries in the table:

x	$\cos(\frac{1}{x})$
± 0.01	0.86232
± 0.001	0.56238
± 0.0001	-0.95216
± 0.00001	-0.99936
± 0.000001	
± 0.0000001	
± 0.00000001	

   Does this further support your answer to question 1?

3. Graph $\cos(\frac{1}{x})$ using desmos or a graphing calculator and adjust the viewing window to zoom in near $x = 0$. Describe the behavior of the function as $x \to 0$. Does the limit $\lim_{x\to 0}\cos(\frac{1}{x})$ exist?

4. Explain why the table in question 1 was misleading.

Example 2

5. Consider the function

$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

   (a) Spend a few minutes attempting to make a table of values to estimate $\lim_{x\to 0} f(x)$ and explain why this method is doomed to fail.

   (b) Spend a few minutes attempting to sketch the graph of $f(x)$ to estimate $\lim_{x\to 0} f(x)$ and explain why this method is also doomed to fail.

The takeaway: These examples illustrate the limitations of using graphs and tables to make conjectures about limits. We will avoid the difficulties above if our conclusions follow only from properties of real numbers, not pictures or tables which offer incomplete information. Going forward, we will develop analytic techniques to prove when a given limit exists. To do so, we first need a precise definition of a limit.

Preliminary definition of a limit: Let f be a function defined on an open interval containing a, with the possible exception of a itself, and let L be a real number. Then we say that $\lim_{x\to a} f(x) = L$ if we can make $f(x)$ *arbitrarily close* to L by taking x *sufficiently close* to a.

This definition, while intuitive, is too vague. We need to rewrite the phrases *arbitrarily close* and *sufficiently close* in terms of mathematical formulas for distances.

1. Write an expression for the distance between x and a.

2. Write an expression for the distance between $f(x)$ and L.

3. If $\lim_{x\to a} f(x) = L$, then moving x closer to a results in $f(x)$ moving closer to L. To see this, experiment with dragging x closer to a in the animation at https://www.desmos.com/calculator/wyngrjjfoi.

   If we can make $f(x)$ "arbitrarily close" to L, then this means that for *any* positive real number ε, we can make the distance between $f(x)$ and L less than ε. Use your answer to the previous question to write "the distance between $f(x)$ and L is less than ε" as an inequality.

   In order to make the distance between $f(x)$ and L less than ε, we have to take x "sufficiently close" to a. How close must x be to a so that the distance from $f(x)$ to L is less than ε? In general, this depends on the function f, the point a, and the given distance ε. Given f, a, and ε, let $\delta > 0$ be the *greatest* real number such that whenever the distance between x and a is less than δ, the distance between $f(x)$ and L will be less than ε.

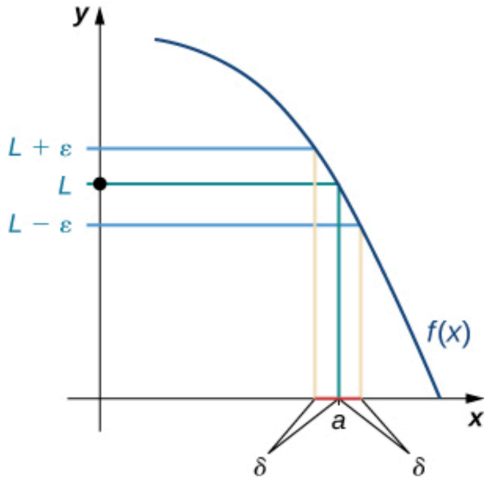
4. Access the animation at https://www.desmos.com/calculator/ufbkxji0xm. Drag the red dashed lines to set ε (the distance from L to either red dashed line). Then, drag the blue dashed lines to set δ (the maximum distance from a to either blue dashed line so that $f(x)$ is between the red dashed lines whenever x is between the blue dashed lines). Write down your values of ε and δ in a table. Now drag the red dashed lines a bit closer together and write the new value of ε. Then drag the blue dashed lines to find the corresponding new value of δ. Repeat this 1-2 more times. Do you see a pattern?

5. Use your answer to question 1 to write "the distance between x and a is less than δ but greater than 0" as a compound inequality. (We have to add "but greater than 0" since $x \neq a$ as $f(a)$ need not be defined.)

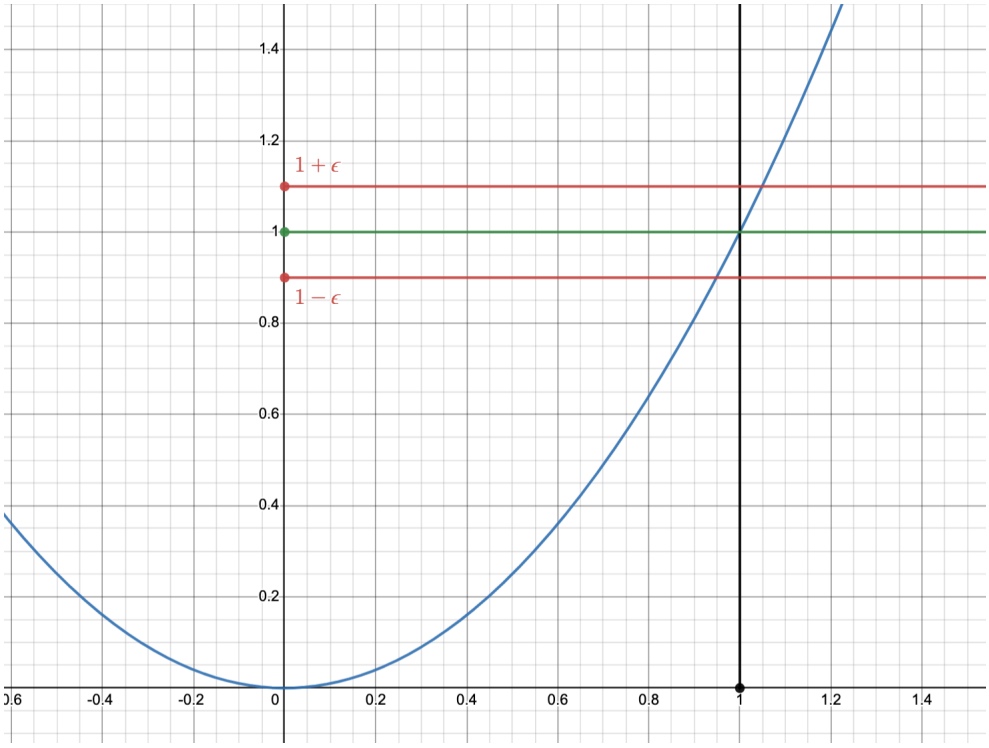
6. Use your answers to questions 3 and 4 to rewrite the definition of a limit more precisely. Finish the sentence: $\lim_{x\to a} f(x) = L$ means that, given any positive distance ε, ...

Precise Definition of a Limit: $\lim_{x\to a} f(x) = L$ means that, given $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that $|f(x) - L| < \varepsilon$ whenever $0 < |x - a| < \delta$.

Another way of saying this is that for any $\varepsilon > 0$, there exists a corresponding $\delta > 0$ such that $f(x)$ is in the interval $(L-\varepsilon, L+\varepsilon)$ whenever x is in the deleted interval $(a-\delta, a) \cup (a, a+\delta)$.



7. For the function f whose graph is shown below, $\lim_{x\to 1} f(x) = 1$. Determine ε from the two horizontal lines drawn below. Then draw two vertical lines so that the graph of $f(x)$ is between the two horizontal lines whenever x is between the two vertical lines. Use your two vertical lines to estimate δ.



8. Access the animation at https://www.desmos.com/calculator/usrnxckvdu, which shows the graph of the function

$$f(x) = \begin{cases} 1 & x \geq 2, \\ -1 & x < 2. \end{cases}$$

(a) The default setting has $L = 1$ and $\varepsilon = 3$. Can you find at least one value of δ so that $f(x)$ is between the two red horizontal dashed lines whenever x is between the two blue vertical dashed lines?

(b) Does your answer to (a) imply that $\lim_{x\to 2} f(x) = 1$? Explain why or why not?

(c) Now drag the red dashed lines closer together so that $\varepsilon = 0.5$. Can you find at least one value of δ so that $f(x)$ is between the two red horizontal dashed lines whenever x is between the two blue vertical dashed lines?

(d) Does $\lim_{x\to 2} f(x) = 1$? Explain why or why not.

(e) Now move L to $L = -1$ and repeat. Does $\lim_{x\to 2} f(x) = -1$? Explain why or why not.

(f) Does $\lim_{x\to 2} f(x)$ exist? Explain why or why not.

♢ **Check your answers with your instructor when you are finished.**

Calculating Limits Analytically - Two Simple Limits

We now use the definition above to *prove* various limits exist. We begin with the two simplest limits.

1. Let c be a real number, and consider the constant function $f(x) = c$. We will prove that, if a is any real number, then $\lim_{x\to a} c = c$.

   (a) Using the animation at https://www.desmos.com/calculator/0e7jqsgfmr, use the precise limit definition to explain why $\lim_{x\to 2} 2 = 2$. That is, given any ε, explain how to find at least one $\delta > 0$ such that $f(x)$ is between the red horizontal dashed lines whenever x is between the blue vertical dashed lines.

   (b) Now write a proof that $\lim_{x\to a} c = c$ by filling in the blanks below:
   Let $\varepsilon >$ _____.
   Taking $f(x) = c$ and $L = c$, $|f(x) - L| =$ _____.
   Choose $\delta =$ _____. Then if $0 < |x - a| < \delta$, then _____ $< \varepsilon$. □

2. Now consider the function $f(x) = x$.

   (a) Using the animation at https://www.desmos.com/calculator/krhys6hi3a, use the precise limit definition to explain why $\lim_{x\to 5} x = 5$. For 3-4 values of ε, find the corresponding value of δ, recording your answers in a table. Do you see any pattern?

   (b) Now write a proof that $\lim_{x\to a} x = a$ by filling in the blanks below:
   Let $\varepsilon >$ _____.
   Taking $f(x) = x$ and $L = a$, $|f(x) - L| =$ _____.
   Choose $\delta =$ _____. Then _____. □

We have proved the following:

Proposition. Let a and c be any real numbers. Then

(a) $\lim_{x\to a} c = c$

(b) $\lim_{x\to a} x = a$.

♢ **Check your answers with your instructor when you are finished.**